

Online Appendix to: Cheap Signals, Costly Proof: Award-Layer Evidence Triage in Cartel Enforcement

Appendix Roadmap

This appendix supplies technical support for the main text and is organized around the evidentiary burden of an evidence-allocation design. [Appendix A](#) documents the award-record universe, the CADE–BEC matching rules, the direct-defendant versus adjudication-anchored cobidder distinction, and the label funnel and sample reconciliation. [Appendix B](#) gives the reduced-form ranking framework, the monotone-likelihood-ratio result, the exit-margin logic, and the platform-survival audit. [Appendix C](#) reports the opportunity-adjusted, timing, leakage, and case-composition audits that discipline alternative interpretations of the ranking. [Appendix D](#) summarizes the economic-profile and score diagnostics. [Appendix E](#) documents the bid-layer benchmark and the cost-recall frontier for sequential award→bid gatekeeping. [Appendix F](#) reports the price-scope evidence and the adaptive-deployment protocol. Full robustness grids, permutation draws, threshold sweeps, the bid-feature dictionary, and additional diagnostic figures are collected in the Online Supplement, whose manifest maps each item to its generating script. Throughout, the appendix interprets the ranking as forensic priority, not proof of conduct or cartel membership, and treats price evidence as scope information rather than damages.

Appendix A. Data Construction and Validation Labels

This appendix records the construction choices behind the award-layer screen and the adjudication-anchored labels, and keeps each source in its proper evidentiary role. The screen is built from procurement records; the validation labels come from CADE adjudications and their BEC co-bidding environments. The design therefore tests whether

a cheap award-layer statistic ranks firms near adjudicated defendants, while leaving legal membership to the case record. The labels are adjudication-anchored exposure measures, not proof of conduct or cartel membership.

Appendix A.1. Award-Record Universe

The BEC procurement universe covers 2009–2019 and contains 1,654,447 tender-item records before the final price-regression restriction; the main price sample contains 1,654,401 observations. For the screening exercise the unit is the firm: a firm enters the outer screening universe if it appears as a participant in at least one BEC tender during the sample window, giving 41,444 participating firms.

Appendix A.2. Screen Construction

For each firm i we construct two award-layer variables, T_i (number of BEC tender-items in which firm i participates) and W_i (number of BEC tender-items won by firm i). The always-loser stratum is the set $\{i : W_i = 0\}$ and contains 16,843 firms. The continuous screening score is $s_i = \log(1 + T_i)$ within that stratum, and the discrete frequent-loser flag is

$$FL_i = \mathbf{1}\{W_i = 0, T_i \geq 14\}.$$

The threshold comes from the always-loser participation distribution—median plus 1.5 times the interquartile range, with statistical cutoff 13.5 and integer rule $T_i \geq 14$ —and is fixed from that distribution alone, not estimated from CADE labels. It flags 2,735 firms, or 16.2% of the always-loser stratum. Because the threshold is pre-specified, the validation exercise asks a genuine question: whether an award-layer rule concentrates firms later observed near adjudicated cartel defendants.

Table A.1: Screen Construction

Object	Value	Construction role
BEC participants	41,444	Outer firm universe for award-record construction
Always-losers	16,843	Zero-win stratum in which the screen ranks firms
Participation threshold	14	Integer rule from median +1.5×IQR
Frequent losers	2,735	Discrete first-stage flag within always-losers
Continuous score	$\log(1 + T_i)$	Rank used when a threshold is not required

Appendix A.3. Matching CADE Defendants and Constructing Cobidder Labels

The starting CADE portfolio contains 65 firm-defendants across 12 adjudicated procurement-cartel cases linked to BEC subject matter. We match CADE defendants to BEC participant records by exact string equality on the full 14-digit CNPJ (*código-fornecedor*), zero-padded to preserve leading zeros; no fuzzy, name-based, or manual matching enters the direct-defendant set. Of the 65 firm-defendants, 47 have at least one BEC tender-item record during the sample window and define the direct-defendant set; the remainder operate outside BEC or lack matchable identifiers.

Direct defendants are not the screen’s target; matching them to BEC tender-items locates the adjudication-anchored exposure set (52013 defendant tender-items, of 48 distinct crossmatch CNPJ of which 41 are `firm_tender_map-active`). The main validation target is the *always-loser cobidder*, constructed from the current scripts by a single rule: a unique always-loser firm (win rate = 0 over 2009–2019) that shares at least one BEC tender-item with at least one BEC-active direct CADE defendant. Three exclusions fix the object. Direct defendants are themselves excluded from the cobidder set, so the label measures loser-side exposure around the anchors rather than the anchors; the sentinel firm code -1 (unidentified bidder) is excluded; and firms appearing across cases or under duplicate CNPJ are counted once. The rule yields 651 always-loser cobidders over the full 12-case portfolio.

The frequent-loser flag is not used to construct the label. It is the award-layer score that is evaluated *against* the label. Of the 651 positives, 341 are frequent losers and 310 are not; the composition is descriptive, not a label restriction. The label is adjudication-anchored exposure, not legal membership: it says the firm appears on the loser side of tenders that include adjudicated defendants, not that it was a defendant or submitted a cover bid in any particular tender.

Appendix A.4. Conservative and Timing Benchmarks

The conservative validation benchmark restricts CADE cases to adjudications with judgment date on or before 2020-12-31, sacrificing sample size to avoid using later case resolution to define the target. Under the *same* cobidder definition, this leaves 4 cases, 19 crossmatch direct defendants, and 208 always-loser cobidders, of which 107 are frequent losers. The cut uses judgment dates because conduct-onset dates are unavailable: only

judgment dates are observed, for 9 of the 12 cases, and the 3 undated cases are placed last in any timing ordering. Per-case timing and benchmark membership are reported in the Online Supplement.

The temporal-holdout exercise computes participation scores from 2009–2016 and evaluates discrimination against labels realized in 2017–2019. The timing-benchmark labels are *retrospective adjudication anchors*: the judgments post-date the bidding, and we claim no real-time legal knowledge at the scoring date. The split follows the enforcement logic rather than a statistical convention: at the moment an agency uses the award layer it does not know the completed case file, and the relevant question is whether earlier participation helps rank firms whose adjudication-anchored exposure is later revealed. Among the 651 positives, 498 are rankable incumbents (some pre-2017 participation history) and 153 are entrant positives with no pre-window history to rank (23.5% of the target). The design is conservative in computing the score before the target period but imperfect because adjudications often post-date conduct that can span several years; it is therefore a retrospective out-of-time discrimination exercise, not causal prediction or a field deployment.

Appendix A.5. Validation Labels and Sample Reconciliation

Table A.2: Validation Labels

Label	Count	Interpretation
Direct CADE defendants in BEC	47	Adjudicated defendants from procurement-cartel cases (legal anchors)
Always-loser cobidders	651	Always-losers (win rate = 0) sharing a tender-item with a direct defendant; main adjudication-anchored exposure target (not membership)
of which frequent losers	341	Composition of the positives (the FL flag does not define the label)
of which non-frequent-losers	310	Composition of the positives
Conservative cases	4	Pre-2020 adjudication benchmark
Conservative direct defendants	19	Crossmatch defendants in the conservative benchmark
Conservative cobidders	208	Always-loser cobidders in the conservative benchmark (same definition)
Conservative frequent-loser cobidders	107	Frequent losers inside the conservative bidder target

The direct-defendant label is kept separate from the cobidder label: the former is a legal anchor, the latter an exposure label tailored to the loser-side ranking. The main validation target and every benchmark in this paper use the *same* cobidder definition; the counts differ only because the candidate pool, the case window, or a common-support restriction differs. The 651 main target applies the rule across the full 12-case portfolio. The 208 conservative count applies the identical rule to the 4 cases judged on or before 2020-12-31, so it is a subset of the main target ($208 < 651$ always-loser cobidders; $107 < 341$ frequent-loser cobidders). No paradox arises: a stricter case window can only remove cobidders, never add them. Table 2 reports all benchmarks side by side, and Table A.3 traces how the positive count moves across exercises as the candidate pool changes while the definition is held fixed.

Two further distinctions matter. The bid-feature common-support pool of 16,731 firms is an evaluation sample on which the Imhof-style bid moments are complete; all 651 positives are retained in it, so it is not a label-funnel count and must not be read as a cobidder or defendant tally, whose exact rows are those of Table 2. Finally, the award-layer screen itself uses no bid amounts, bid ranks, within-tender dispersion, reference prices, LANCES timestamps, CADE outcomes, or price-regression residuals; those enter later as validation or forensic benchmarks, keeping the first-stage statistic computable in award-only environments and the bid layer available as an independent forensic stage.

The positive-count concordance in Table A.3 shows that the 651 positives are stable across exercises under one fixed definition; the counts that differ (498 rankable incumbents, 153 entrant positives, 208 conservative) are common-support, timing, or case-window subsets, not alternative label definitions.

Appendix B. Evidence-Allocation Framework and Legal Scope

Appendix B.1. The pre-proof ranking problem

The model is intentionally spare: it justifies an ordering rule for scarce forensic attention, not the internal organization of a cartel (formation, side payments, leniency, and damages are treated in Asker, 2010, and Marshall and Marx, 2012). The narrower object is the pre-proof enforcement problem—why a firm that repeatedly loses while still appearing in procurement records can be a useful target for bid-level follow-up. The device is a reduced-form evidence-allocation rule.

Table A.3: Positive-count concordance across validation exercises.

Analysis	Candidate pool	Support restriction	Pos.	Pool N	Why the count differs
Main label funnel (Table 2)	Always-losers (16,843)	none	651	16,843	Reference definition; full 12-case portfolio
Opportunity-adjusted validation	Exposed always-losers (common support)	exposure common support (E_i defined)	651	6,040	Exposed subsample retains all positives; denominators shrink to common support
Strict timing, full test universe	All BEC firms scorable at screening date	score frozen 2009–2016; entrants at score zero	651	41,444	Universe widens to every scorable firm; 153 positives are unrankable entrants
Strict timing, rankable incumbents	Firms with pre-2017 history	pre-window participation > 0	498	32,682	Entrant positives (153) excluded: no pre-window history to rank
Strict timing, training always-loser pool	Always-losers in training window	train-window AL status	651	21,819	Pool defined by train-window zero-win status
Bid-feature common support	Always-losers with complete Imhof moments	complete bid-feature support	651	16,731	112 AL firms lack priced-bid moments; all positives retained
Cost-recall frontier	Bid-feature common-support pool	same as bid benchmark	651	16,731	Same pool; cost denominators added
Conservative pre-2020 benchmark	Always-losers (16,843)	case-window restriction only	208	16,843	Fewer anchor cases under the SAME definition (subset of 651)

Notes. The target definition is identical in every row (broad always-loser cobidder; direct defendants excluded; frequent-loser status never used). Positive counts differ only because later exercises impose common-support, timing, or bid-feature restrictions on the candidate pool.

In a cartel-targeted tender one firm is assigned the winning role and submits the designated bid b^* ; a losing-role participant submits $b_\ell > b^*$ and is not meant to win. Winning by deviation creates an internal sanction $\kappa > 0$ (loss of future rents, relational standing, or capacity-allocation rights); let r be the private gain from deviating into the winner role. Firms outside the winning role are of two reduced-form *model* types: type C , cartel-deployed losing-role participants, and type G , ordinary losing participants (inefficient, capacity constrained, or unlucky). The labels are a device for stating the ranking condition; C and G are not observed classes. The empirical screen conditions on the observed event $W_i = 0$ (W_i is the firm’s number of wins) and ranks firms by participation count T_i .

Lemma 1 (Zero-win losing role). *If the internal sanction from winning in a losing role exceeds the private deviation gain, $\kappa > r$, a losing-role participant avoids winning: it submits $b_\ell > b^*$, so $\Pr(W_i > 0 \mid C) = 0$ over tenders in which the firm is deployed in the losing role.*

Proof. Submitting $b_\ell > b^*$ yields the deployment payoff net of the role cost. Deviating to a winning bid gains r but pays κ , which is unprofitable when $\kappa > r$. The inequality fixes the win probability of the losing role, not the numerical bid. $\square$

The inequality $\kappa > r$ supports conditioning on the zero-win event but does *not* imply that all zero-win firms occupy a losing role: the conditioning set is intentionally broad,

since many ordinary firms also never win. The evidentiary content comes from the second observable—participation intensity—and from the dynamic margin governing who remains in the zero-win set.

Appendix B.2. Dynamic participation, exit, and the ranking statistic

The ranking statistic is informative only if persistent zero-win participation is more characteristic of a deployed losing role than of an ordinary losing firm. We obtain that property *endogenously* from a participation-and-exit margin rather than assuming a type-specific rate gap. An ordinary zero-win firm G pays a participation cost $c_G > 0$ and earns expected gross surplus $m_G(H_{it})$, with H_{it} its loss history at item i , time t ; under Bayesian updating, repeated losses weakly lower m_G . It continues iff

$$V_G(H_{it}) = m_G(H_{it}) - c_G + \beta \mathbb{E}[V_G(H_{i,t+1}) \mid H_{it}] \geq 0, \quad (\text{B.1})$$

and *exits* when $V_G(H_{it}) < 0$. A cartel-deployed losing-role firm C instead values continuation through a role value $a_C(H_{it})$ (future rents, relational or subcontracting value, internal compensation for sustained losing). It continues iff

$$V_C(H_{it}) = a_C(H_{it}) - c_C + \beta \mathbb{E}[V_C(H_{i,t+1}) \mid H_{it}] \geq 0. \quad (\text{B.2})$$

Because a_C derives from the role rather than the firm's own win prospect, it does not erode with the loss count the way m_G does, so after sufficiently many zero-win periods

$$a_C(H_{it}) - c_C > m_G(H_{it}) - c_G, \quad (\text{B.3})$$

and C keeps participating while G is selectively removed. The empirical consequence is a *selection* result: among surviving zero-win firms,

$$\lambda_C > \lambda_G \quad \text{on the surviving conditioning set } W_i = 0, \quad (\text{B.4})$$

derived from differential exit under (B.1)–(B.3), not posited as an exogenous type effect.

Assumption 2 (Role-value persistence). *Among surviving zero-win firms, cartel-deployed losing-role participants have weakly greater net continuation value from repeated*

losing than ordinary zero-win firms after sufficiently many losses: there is a loss count $\bar{L}$ such that for histories with accumulated losses exceeding $\bar{L}$, $a_C(H_{it}) - c_C \geq m_G(H_{it}) - c_G$.

Conditional on survival and zero wins, let participation counts follow a Poisson approximation,

$$T_i \mid \tau_i, q \sim \text{Poisson}(\lambda_q \tau_i), \quad q \in \{C, G\}, \quad (\text{B.5})$$

with τ_i the length of time firm i is observed; (B.4) supplies the ordering $\lambda_C > \lambda_G$ on this surviving set.

Proposition 3 (Persistence ranking). *Suppose (1) losing-role firms avoid winning when $\kappa > r$ (Lemma 1); (2) ordinary zero-win firms exit when continuation value falls below the outside option ((B.1)); (3) role value keeps C in the risk set longer, so the selection wedge (B.3)–(B.4) holds (Assumption 2); and (4) among survivors effective participation intensity is higher for C than for G , $\lambda_C > \lambda_G$. Then the posterior $\Pr(C \mid T_i = t, W_i = 0, \tau_i)$ is non-decreasing in t , and T_i and $\log(1 + T_i)$ are monotone ranking statistics for forensic priority within the zero-win stratum.*

Proof. By Bayes’ rule the posterior odds of C against G are proportional to the Poisson likelihood ratio $\exp[-(\lambda_C - \lambda_G)\tau_i](\lambda_C/\lambda_G)^t$, which under $\lambda_C > \lambda_G$ is increasing in t (monotone likelihood ratio of the Poisson family; Karlin and Rubin, 1956).¹ The posterior is non-decreasing in the count, and $\log(1 + T_i)$ preserves the ranking. $\square$

Condition (4)—the surviving participation gap—is the key empirical content, validated against adjudication-anchored labels in §§4–6. This is a ranking result under maintained behavioral conditions, not an identification claim. The main-text binary rule (flagging firms above the median plus 1.5 times the interquartile range of zero-win participation) is an operational coarsening of this ranking; the threshold is not a structural parameter, and agencies could instead use the continuous score, a top- k list, or a budget-calibrated cutoff.

Corollary 4 (Legal scope). *Under Lemma 1, Assumption 2, and Proposition 3, the frequent-loser statistic orders expected investigative value within the zero-win stratum. It is not sufficient for liability, damages, or cartel membership.*

¹Karlin, S. and H. Rubin (1956). The theory of decision procedures for distributions with monotone likelihood ratio. *Annals of Mathematical Statistics* 27(2), 272–299.

Proof. Triage requires an ordering of expected investigative value, which Proposition 3 supplies. Liability and damages require different predicates—proof of agreement, conduct in particular tenders, and a measure of price injury—that the statistic, carrying only participation and win information, omits. $\square$

Appendix B.3. Empirical survival audit

The exit margin of §Appendix B.2 predicts who remains on the platform: if role value sustains cartel-deployed losing participation while ordinary zero-win firms are selectively removed, the most exposed firms should display longer continued participation. We audit this descriptively on a firm-year panel of the 16,843 always-losers (101,366 firm-years, 2009–2019, right-censored at 2019). Because the audit uses *full-sample* labels, it is retrospective and mechanism-supporting at most; it does not test cartel membership or conduct. “Survival” means continued BEC *participation*, not legal-entity survival.

The pattern is in the predicted direction. Table B.1 shows that FL cobidders persist markedly longer: mean active years 3.51 versus 2.87 for FL non-cobidders and 1.61 for non-FL always-losers; median duration 3 versus 2 versus 1 years; exit share before 2019 of 0.63 versus 0.73 versus 0.85. A discrete-time hazard model (cloglog, duration and calendar-year fixed effects, clustered by firm) confirms the ordering, with exit-hazard ratios of 0.231 for FL cobidders and 0.365 for FL non-cobidders relative to non-FL always-losers—both far below one—so exposure to the labels is associated with substantially lower exit hazard (full hazard estimates, the Cox specification, and the sensitivity battery are reported in the Online Supplement). The cobidder coefficient is sharply estimated ($z = -22.2$, $p < 10^{-100}$), and the Kaplan–Meier curves are clearly separated. The direction is stable across robustness variants—alternative exit definitions, excluding late cohorts, and excluding the largest CADE case (ratio 0.281)—and a timing-disciplined to-date label attenuates it to 0.768 but leaves it below one.

Appendix B.4. Evidentiary scope

The survival result is consistent with the exit-margin logic—role value sustaining persistent losing participation—but is mechanism-supporting at most; ordinary zero-win firms may exit for many reasons. More generally, even if the model is correct, a high value of $\log(1 + T_i)$ raises the posterior probability of loser-side exposure only relative to

Table B.1: Platform survival of always-losers, by full-sample label.

Group	<i>N</i> firms	Mean active years	Exit share < 2019	Median duration	Right-cens. share	Hazard ratio
Non-FL always-loser	14,104	1.61	0.85	1	0.15	1.00
FL non-cobidder	2,391	2.87	0.73	2	0.27	0.365
FL cobidder	341	3.51	0.63	3	0.37	0.231

Firm-year panel of 16,843 always-losers, 101,366 firm-years, 2009–2019, right-censored at 2019. “Active years” counts distinct years of platform participation; “duration” is years between first and last appearance. Hazard ratios are adjusted discrete-time (cloglog) estimates relative to the non-FL always-loser baseline (full hazard table in the Online Supplement). Labels are full-sample and the audit is therefore descriptive and retrospective: it is mechanism-supporting at most and is not a test of cartel membership.



Figure B.1: Kaplan–Meier platform-survival curves for always-losers, by full-sample label.

Alt text: Three Kaplan–Meier curves of the share of firms still participating on the platform against tenure in years. The FL-cobidder curve lies highest and declines most slowly; the FL-non-cobidder curve is intermediate; the non-FL always-loser curve falls steepest and lies lowest, with most firms gone after one year.

Note: Right-censored at 2019. Labels are full-sample, so the comparison is descriptive. “Survival” is continued BEC platform participation, not legal-entity survival.

other zero-win firms: it does not identify an agreement, assign cartel membership, prove any specific bid was cover, or establish a damages base. Those claims require the richer record (bid values, communications, leniency material, winner-rotation patterns). This boundary is the legal reason for the staged architecture in the main text: the award layer ranks firms on a statistic that is cheap, observable, and monotone under a transparent behavioral assumption, while the bid layer and case record do the evidentiary work that liability requires. The paper’s identification continues to rest on the opportunity-adjusted and timing-disciplined tests of §§4–6.

Appendix C. Validation Audits

The validation design addresses three mechanical channels. Frequent participants have more chances to appear near any defendant, so raw co-bidding rates are not enough; firms may operate in the same procurement environments as CADE anchors, giving them more opportunities to meet those anchors even under ordinary competition; and item-level labels can mechanically reuse the same participation history that constructs the score. This appendix records the opportunity and timing audit logic and the interpretation each audit licenses; the main text reports the headline results.

Throughout, “CADE-defendant contact” denotes an *adjudication-anchored exposure* label: a firm shares a tender-item with a directly adjudicated CADE defendant. It does not assert that the firm is a cartel member, party to any agreement, or liable for anything. The audits validate forensic priority, not membership, damages, or proof. Online Supplement C reports the full grids behind every audit summarized here.

Appendix C.1. Opportunity-Cell Construction and Expected Contact

Purpose. A firm can appear near a CADE anchor simply because it operates in environments where defendants are active; the opportunity cells discipline this exposure before any score is credited.

Method. We partition the award layer into procurement-opportunity cells g and benchmark each firm against others facing the same cells, using three nested definitions: **COARSE** = item group (two-digit item-code prefix) $\times$ year; **MEDIUM** = item group $\times$ year $\times$ buyer/procuring-unit (the reference definition for the headline within-stratum results); and **STRICT** = item code $\times$ year $\times$ buyer. Modality (Convite vs. Pregão) is

intentionally dropped: it is not present in the award layer used here, and conditioning on it would further fragment already-sparse cells (revisited as a sensitivity below). For firm i , with n_{ig} the tender-items held in cell g and p_g the cell defendant-contact rate, *expected* contact under ordinary opportunity is $E_i = \sum_g n_{ig} p_g$, the *observed* contact O_i is the firm’s actual count of defendant-sharing tender-items, and the *excess* is $X_i = O_i - E_i$ with standardized form $Z_i = (O_i - E_i) / \sqrt{\sum_g n_{ig} p_g (1 - p_g)}$. To prevent a firm from being benchmarked against itself, p_g is computed leave-one-out.

Result. The mean cell defendant-contact rate $\bar{p}_g$ is around 0.017–0.019 across the three definitions; finer cells gain resolution at the cost of support (under STRICT, 99.5% of cells hold fewer than five tender-items and 75.7% are singletons). Excess concentrates where the score is high (mean $X_i = +0.068$ at participation tier $T \geq 50$, versus ≈ -0.006 to -0.000 at lower tiers), but most of the level of O_i is explained by E_i : exposure does the heavy lifting and excess is the residual the score must rank. Common-support retention is honest about sparsity — under MEDIUM cells the strict support rule retains about 46% of firms (the within-stratum results use MEDIUM), and under STRICT retention collapses to about 9%. The cell-construction, expected-contact, and retention grids are in Online Supplement C.

Appendix C.2. Opportunity-Adjusted Validation

Purpose. The control-function audit asks whether the screening score still ranks defendant contact once opportunity exposure E_i is in the model.

Method. We estimate a ladder of logits and rank-tests (S0–S6) on the always-loser firms, reported in Table C.1: raw baselines (S0, including an exposure-only model), flexible participation (S1), score-plus-log(1+ E) at each cell granularity (S2), breadth controls (S3), opportunity- and participation-fixed effects (S4), the binary excess target $\mathbf{1}[X_i > 0]$ (S5), and the strict-support subsample (S6).

Result. The honest reading is deflationary. Exposure alone already classifies cobidder contact well (exposure-only AUC 0.905 unconditionally; 0.713 on the exposed subsample), and the combined exposure-plus-score AUCs near 0.98 overstate the score’s marginal value because E_i encodes the label. Once exposure is held fixed, the residual ordering is *marginal at best and not robust across designs*: the nested DeLong increment is only +0.010 ($p = 0.013$), the within-opportunity-stratum AUC for the score is 0.471 ($\approx$

Table C.1: Control-Function Validation (Selected Specifications)

Spec	Model	N	Pos.	ROC AUC	PR-AUC
S0	raw score ($\log T$)	16,843	651	0.761	0.143
S0	raw FL14	16,843	651	0.688	0.097
S0	exposure-only (all AL)	16,843	651	0.905	0.300
S2	score + $\log(1+E)$ [MEDIUM]	16,843	651	0.985	0.708
S2	FL14 + $\log(1+E)$ [MEDIUM]	16,843	651	0.985	0.706
S2	exposure-only logit (exposed)	6,040	651	0.713	0.300
S3	score + E + breadth	16,843	651	0.983	0.711
S4	score + opp-FE + part-FE (exposed)	6,040	651	0.707	0.255
S5	score ranks $\mathbf{1}[X_i > 0]$ (MEDIUM)	16,843	475	0.764	0.095
S5	FL14 ranks $\mathbf{1}[X_i > 0]$ (MEDIUM)	16,843	475	0.687	0.073
S6	score + E , strict support	7,683	446	0.969	0.671
NESTED	exposure-only logit	6,040	651	0.713	—
NESTED	exposure + score	6,040	651	0.723	—
WITHIN	score opportunity stratum	6,040	651	0.471	—
WITHIN	FL14 opportunity stratum	6,040	651	0.507	—

Notes: Outcome is adjudication-anchored cobidder contact (S5 uses the binary excess target $\mathbf{1}[X_i > 0]$). **The combined exposure-plus-score logits in S2–S3 reach AUC ≈ 0.98 , but this is mechanically inflated and is NOT evidence of score signal:** the exposure term E_i is built from the same defendant-contact rates that define the label, so a model containing E_i partly predicts the label by construction. The score’s *isolated* contribution is not read off these combined AUCs. Holding exposure fixed, the residual ordering is marginal at best: the nested DeLong increment is only +0.010 over the exposure-only model ($p = 0.013$), and the within-opportunity-stratum AUC for the score is 0.471 ($\approx$ chance; FL14 0.507). The companion permutation and matched audits in Online Supplement C do not establish a robust residual.

chance), and the FL14 binary is no better (0.507). The score does rank the binary excess-contact target at AUC 0.764 within MEDIUM cells, but the three companion exercises in Online Supplement C do not corroborate a robust residual signal: a matched-stratum label permutation places the observed PR-AUC at $p = 0.127$ (not significant), a firm-exposure simulation out-predicts the raw score on level PR-AUC ($p = 1.000$), the CEM opportunity-matched comparison gives a within-stratum AUC of 0.626 ($n = 6,039$, full common support), and the within-stratum change in $\Pr(\text{cobidder})$ for FL14 versus non-FL is in fact *negative* (-0.017). We therefore report no robust residual ordering net of opportunity; the only marginally positive quantity is the small nested increment, which the permutation tests do not support as reliable.

Intensity-restricted label sensitivity. A natural objection is that the broad shared-tender-item label dilutes genuine exposure with incidental single-contact cobidders, masking a residual signal. We re-ran the full opportunity-adjusted battery restricting positives to cobidders sharing at least two distinct defendant tender-items (368 of the 651 positives). The restriction strengthens the deflationary conclusion rather than reversing

it: raw discrimination rises mechanically (ROC-AUC 0.854), but the exposure-only model rises faster (0.873 on the exposed subsample; 0.956 unconditionally) and again out-predicts the score; the within-opportunity-stratum AUC remains at chance (0.506); the nested increment falls to +0.003 ($p = 0.47$); and the matched-stratum permutation again does not reject ($p = 0.39$). The more tightly the label is anchored to intense defendant contact, the more completely procurement opportunity accounts for it.

Appendix C.3. Timing Information Sets and Strict Holdout

Purpose. This block holds *time* fixed: could a regulator who froze the score *before* the evaluation window have ranked the firms later adjudicated as cobidders, without peeking at future participation, wins, threshold, or CADE adjudications?

Method. The CADE files carry only judgment dates (`data_julgamento`, mostly 2015–2025) and *no conduct dates*, so all timing is *year-level* (BEC participation year), never day-level (`DAY_LEVEL_TIMING_UNAVAILABLE`). We define seven candidate information sets, A–G, distinguished by which inputs (score, zero-win status, threshold, label) are drawn pre- versus post-window. We lead on the honest incumbent-triage designs — C/D/G (strict prospective score, rankable incumbent pool, retrospective label, read as triage rather than prospective deployment) — report A as a labeled retrospective upper bound, B as a leak-flagged decomposition, E to disclose the entrant coverage hole, and document F (adjudication-observable-at-time) as infeasible because all judgment dates post-date a 2009–2016 freeze. The strict holdout freezes the score, always-loser status, and median+1.5×IQR threshold on 2009–2016 (train-window threshold 7, vs. full-window 13.5) and evaluates against 2017–2019 cobidder labels. Table C.2 gives five nested samples; the full A–G information-set register and the leakage-channel cross-tabulation are in Online Supplement C.

Result. The reading is deliberately deflationary. A within-pool ranking signal survives (sample 3, AUC 0.684–0.646), but on the full firm universe the strict screen has no operational traction (precision@500 = 0): the entrant coverage hole and the mass of score-zero ties are the proximate causes. Two companion exercises in Online Supplement C confirm the pattern. A rolling-origin validation expanding the training window for $t = 2014, \dots, 2019$ posts PR-AUC ≈ 0.012 –0.014 and precision@500 = recall@500 = 0 on the full universe in every origin year, with the worst ROC below chance (0.446, 2015);

Table C.2: Strict Holdout 2009–2016 → 2017–2019 (Year-Level Timing)

Sample	N	Pos.	ROC AUC	PR-AUC	Prec@500	Recall@500	Lift@500	Tie-at-0
1. Full test candidate	41,444	651	0.474	0.014	0.000	0.000	0.00	0.246
2. Rankable ($T^{\text{train}} > 0$)	32,682	498	0.471	0.013	0.000	0.000	0.00	0.043
3. Training always-loser	21,819	651	0.684	0.077	0.124	0.095	4.16	0.402
4. Training AL & test-active	11,351	312	0.667	0.084	0.142	0.228	5.17	0.772
5. Zero-win both (LEAKS)	16,843	651	0.646	0.086	0.136	0.104	3.52	0.280

Notes: Continuous loser-side score ($\log T^{\text{train}}$). Sample 3 is the honest strict pool; within it, the FL-binary AUC is 0.646 and the continuous AUC is 0.684 (threshold = 7, vs. full-window 13.5). Sample 5 leaks future wins via full-period always-loser status and is reported as a leak-flagged sensitivity only. **On the full candidate universe (sample 1) the strict screen collapses to chance:** ROC 0.474 (below chance), PR-AUC 0.014, and precision@500 = recall@500 = 0, with about 25% of firms tied at score zero. Of the 498 rankable positives and 153 entrant positives (23.5% of 651), entrants form a structural blind spot the prospective score cannot reach. “Tie-at-0” is the share of the candidate set carrying score zero. Source: `table_D_strict_2009_2016_to_2017_2019.csv`; `strict_holdout_composition.csv`.

the within-pool (training-always-loser) ROC stays in the 0.66–0.70 range across origins, again an artifact of a thicker retrospective training history rather than real-time learning. A leakage decomposition of the in-sample item-level reference AUC (0.995) confirms that the surviving structural component stays in the 0.86–0.89 range under cobidder-firm holdout (out-of-fold AUC 0.891) and temporal holdout (AUC 0.864), with the 0.10–0.13 AUC lost to mechanical reuse disclosed as such.

Appendix C.4. Case Composition

Purpose. A generalization concern is whether the pooled signal is carried by one or two adjudicated cases rather than the construct.

Method. We link six cases with enough cobidders to score (a seventh, `sacos_de_lixo`, is too sparse and is flagged) and re-evaluate the ranking after dropping the largest and the top-two cases, and under case-balanced weighting. Table C.3 reports the leave-largest-case-out collapse.

Table C.3: Case Dominance: Leave-Largest-Case-Out

Scenario	Pos.	ROC AUC	PR-AUC	Prec@500	Rec@500
Full (pooled)	651	0.761	0.143	0.216	0.166
Drop largest case	443	0.763	0.090	0.132	0.149
Drop top-two cases	268	0.752	0.058	0.092	0.172
Case-balanced (mean over cases)	—	—	—	0.043	0.173

Notes: Dropping the single largest case (`trens_metros`) lowers PR-AUC from 0.143 to 0.090 (−37%) and precision@500 from 0.216 to 0.132; dropping the top two cases takes PR-AUC to 0.058. **Case-balanced precision@500 (weighting each case equally rather than each firm) is 0.043, against the firm-weighted pooled 0.216:** the pooled operational number is concentrated in the largest case. ROC barely moves (it is uninformative at this prevalence). Source: `table_H_case_dominance_validation.csv`.

Result. Dropping the single largest case lowers PR-AUC from 0.143 to 0.090 (−37%);

case-balanced precision@500 is 0.043 against the firm-weighted pooled 0.216, so the pooled operational number is concentrated in the largest case (which holds 32.0% of positives and 45.4% of the true positives in the top 500). The concentration is structural across the environment, documented in Online Supplement C: the largest item group holds 21.5% of positives (item-group HHI 0.100) while the buyer distribution is diffuse (buyer HHI 0.009), and dropping the largest item group leaves PR-AUC essentially unchanged ($0.143 \rightarrow 0.134$). A clustered randomization-inference test permuting cobidder labels within item-group $\times$ year clusters ($B = 1,000$) clears the null on the ranking metrics (ROC, PR-AUC, precision@500 all $p = 0.001$), but top-500 case coverage is *not* significant ($p = 0.103$), consistent with one case anchoring the operational signal.

Appendix C.5. Direct-Defendant Scope Check

If the statistic were a generic cartel-membership detector, it should classify direct CADE defendants as well as cobidders. It does not: the FL-binary score (retrospective incumbent triage) ranks direct defendants at $\text{ROC} \approx 0.49$ under both full and strict timing (0.491 and 0.492), i.e. at chance, with $\text{precision@500} = 0$. Continuous participation volume does carry some information about defendants—ROC 0.66–0.70, modestly above chance—but with $\text{precision@500} \leq 0.004$, so it does not operationally retrieve them either (full grid in Online Supplement C). The asymmetry is the design: the loser-side *binary* flag is uninformative about defendants, who typically sit on the winner side of the arrangement, while participation volume is not entirely silent. This is a scope boundary, not an escape hatch.

Appendix C.6. Limitations

Several limitations bound this appendix. The opportunity cells are *award-layer proxies* built from item, year, and procuring-unit identifiers, not a full description of each firm’s competitive opportunity set, and procurement modality is dropped because it is absent from that layer. All timing is *year-level* (DAY_LEVEL_TIMING_UNAVAILABLE): the CADE files carry only judgment dates (2015–2025), no conduct dates, so we cannot align the screen to when a cartel operated and the adjudication-observable design is infeasible rather than estimated. There is an *entrant coverage hole* (153/651, 23.5%, of positives carry score zero with no training history) and the signal is *concentrated* by case (dropping the

largest cuts PR-AUC by 37%), item group, and year, so the pooled operational number overstates breadth. ROC is a red herring at sub-1% prevalence: precision and recall—not ROC—are the operational metrics, and they stay low under the pooled diagnostic and collapse to zero on the full strict-timing universe. Most importantly, the CADE-defendant labels are *adjudication-anchored exposure* labels, not membership findings: a tagged firm shared a tender-item with an adjudicated defendant; it is not thereby a cartel member, a party to any agreement, or liable for anything. Net of opportunity and timing, the residual ordering is *marginal at best and not robust across designs*; what the award layer reliably delivers is a coarse, retrospective basis for prioritizing scrutiny, and it is **not legal proof, not identification of cartel members, and not an estimate of damages or causal effect**.

Appendix D. Economic Profile and Robustness: Construction and Diagnostics

This appendix documents the construction and diagnostics behind the Section 5 economic-profile and robustness exercises. Throughout, “cobidder” denotes an *adjudication-anchored exposure label* — a firm that co-bid with a firm named in a CADE proceeding — and never a finding of participation in any agreement. The profiles below describe *losing behavior* on the award layer; they are descriptive inputs to a triage ranking, not proof of conduct.

Appendix D.1. Comparison Groups and Profile Variables

All groups are drawn from the always-loser universe (firms with zero recorded wins over the calibration window, $N = 16,843$). The adjudication-anchored validation target is the broad set of 651 always-loser cobidders—firms that share at least one tender-item with a BEC-active direct CADE defendant—of which 341 (52.4%) satisfy the frequent-loser cut and 310 (47.6%) do not, so the label is not frequent-loser-conditioned. The 47 direct CADE defendants are *winners* and are *excluded* from every cobidder group; the 6 that appear in the always-loser universe are excluded from the profile contrasts to keep the comparison free of named-defendant contamination. The Section 5 profile contrast is run *within* the frequent-loser stratum—group D (FL non-cobidders, $N = 2,394$) versus group

E (FL cobidders, $N = 341$)—so that the estimated gaps cannot be an artifact of the cut. Online Supplement D lists all group definitions.

Profile variables are computed from award records over the calibration window and organized in five panels: (A) participation and survival (T_i , active years, items per active year); (B) breadth and concentration (distinct buyers, item groups, modality mix, item-group and buyer HHIs); (C) defendant-proximity / opportunity layer (observed contacts O_i , opportunity-expected contacts E_i^{MED} , excess $X_i^{\text{MED}} = O_i - E_i^{\text{MED}}$, cases touched, contact intensity O_i/T_i , share in defendant-bearing cells); (D) modality-quorum proxies (Convite minimum-quorum share, mean competitors faced); and (E) timing (first-year and late-half participation shares). Three variable classes that would sharpen interpretation are unobserved in the award layer and used nowhere in Section 5: tender value / contract size (price is recorded only for the *winner*), distance-to-winner (requires bid microdata), and fine geography / specialization. Because of the latter two, the ordinary-loser alternatives in Section 5 are *narrowed but not eliminated*.

Appendix D.2. Standardized Differences

Section 5 compares groups D and E using standardized mean differences (Cohen’s d) rather than p -values. With N in the thousands, p -values reject essentially any difference and convey no information about magnitude; the SMD is scale-free and comparable across heterogeneous variables. For a continuous variable with group means $\bar{x}_E, \bar{x}_D$ and standard deviations s_E, s_D ,

$$d = \frac{\bar{x}_E - \bar{x}_D}{s_p}, \quad s_p = \sqrt{\frac{(n_E-1)s_E^2 + (n_D-1)s_D^2}{n_E + n_D - 2}},$$

with the binary analogue obtained by substituting $\hat{p}(1 - \hat{p})$ for the variances. A positive d means cobidders (E) are higher. Table D.1 reports the top-ranked differences; the full variable table is in Online Supplement D. The largest gaps are the defendant-proximity variables (panel C), which are partly mechanical by construction; the non-mechanical breadth and persistence gaps (panels A and B) are moderate, with $|d|$ near 0.4–0.5.

The defendant-proximity gaps in panel C are *partly mechanical*: the cobidder label is defined by adjudication-anchored contact, so firms with more contact are mechanically more likely to be labeled. We hold exposure fixed by regressing each profile variable

Table D.1: Standardized profile differences (top variables), E (FL cobidders) minus D (FL non-cobidders).

Panel	Variable	Label	Raw diff.	Cohen's d
C	n_cases	# CADE cases touched	1.109	6.169
C	n_def_firms	# distinct defendants	1.370	4.270
C	$contact_intensity$	O_i/T_i	0.172	2.119
C	$share_in_def_cells$	part. in def. cells	0.205	1.829
C	O_i	observed contacts	7.658	1.806
C	X_i^{MED}	excess contacts	1.523	1.212
B	n_buyers	# distinct buyers	7.831	0.532
A	T_i	# tender-items	24.689	0.438
B	$pregao_share$	Pregão share	0.173	0.436
A	n_years	# active years	0.625	0.391
B	$convite_share$	Convite share	-0.173	-0.436

Notes: $n_D = 2,394$, $n_E = 341$ (modality shares use $n_D = 2,387$ where modality is defined). Positive d : cobidders are higher. Top eleven variables by $|d|$; the full table is in Online Supplement D. The largest gaps are defendant-proximity variables (n_cases $d = 6.17$, n_def_firms $d = 4.27$), which are partly mechanical because the cobidder label is itself adjudication-anchored. The non-mechanical breadth and persistence gaps are moderate (n_buyers $d = 0.53$, T_i $d = 0.44$, n_years $d = 0.39$). Item-group HHI shows only a small raw gap ($d = 0.13$) that falls to ≈ 0 after opportunity adjustment.

on the cobidder indicator with controls for $\log E_i^{MED}$, a binned score term, breadth, and active years, and by coarsened exact matching on E_i^{MED} deciles. Under either adjustment the item-group HHI gap falls from $d \approx 0.13$ to ≈ 0 , and the breadth (distinct buyers) and persistence (active years) gaps likewise collapse toward zero, so they reflect procurement-environment composition rather than intrinsic firm traits. Only the defendant-contact variables (n_cases , n_def_firms , X_i) survive, and that survival is *partly mechanical* because the label is itself defined by defendant contact; it should not be read as independent corroboration of conduct.

Appendix D.3. Robustness: Monotonicity, Threshold, Markets, and Null Controls

Four robustness families bear on the exposure-ranking reading; full detail is in Online Supplement D, which reports the threshold sweep, the bunching histogram, the market-specific zero-win variants, and the negative controls. *Monotonicity*. Across participation bins, cobidder prevalence rises (Spearman $\rho = +0.99$) while mean excess contact shows no rising pattern ($\rho = -0.25$): the score ranks firms by *exposure*, not collusion intensity. *Threshold*. The FL14 cut $T_i \geq 14$ equals median $+ 1.5 \times \text{IQR}$ ($= 13.5$, rounded up) of the always-loser participation distribution; it is an *administrative* construction, computed ex post and *never published*. Discrimination changes smoothly across a placebo-threshold grid with no discontinuity at 14, and the participation distribution declines smoothly through the cut (local density ratio 1.06, indistinguishable from 1), so there is no strategic

bunching. *Market-specific zero-win.* Four home-market loser definitions overlap the global set on only 0.33–0.55 of their losers; the global definition is not dominated by any of them (global ROC-AUC 0.761 vs. the best home-market alternative 0.584) and is stable to leave-one-item-group-out re-estimation. *Negative controls.* Against matched null targets ($B = 500$ draws, seed 20260603), the real-anchor ROC-AUC (0.761) is statistically indistinguishable from the matched placebo-anchor distribution (mean 0.755; empirical $p = 0.46$), and anchors built from non-CADE high-volume winners generate raw discrimination at least as high (null mean 0.78; $p = 0.91$). The negative controls therefore corroborate the opportunity decomposition rather than rescue the screen: the raw ranking reflects generic co-participation arithmetic, not a signature specific to the adjudicated defendant network.

Appendix D.4. Limitations

The defendant-contact variables carry the largest standardized gaps but are partly a construction artifact, and the non-mechanical breadth and persistence gaps largely collapse once opportunity is held fixed; the score ranks firms by exposure to defendant-adjacent opportunities, not by conduct intensity; ordinary specialization and geographic-niche alternatives are narrowed but not eliminated because distance-to-winner, geography, and tender value are unobserved. Award-layer profiles describe patterns of *losing*; they are not proof of any agreement, and “cobidder” is throughout an adjudication-anchored exposure label, never an assertion of membership.

Appendix E. Bid-Layer Benchmark and Cost-Recall Details

This appendix documents the construction and diagnostics behind the bid-layer benchmark used in Section 6.1, which asks whether the cheap, award-layer participation signal carries ranking information comparable to a transparent bid-moment random-forest benchmark inspired by Imhof–Wallimann-style screens, built from full bid microdata. The bid-layer benchmark is a *forensic-stage, full-observability* diagnostic: it computes seven Imhof–Wallimann within-tender bid-distribution moments, aggregates them to the firm, and feeds them to a random forest evaluated by cross-validation. It is not a state-of-the-art horse race and not a first-stage triage tool, because the moments require all within-tender bids to be observed and recovered. Throughout, “cobidder” denotes

an *adjudication-anchored exposure label* — an always-loser firm that shares at least one BEC tender-item with a BEC-active direct CADE defendant — and never a finding of participation in any agreement; the benchmark *ranks and prioritizes* the adjudication-anchored cobidder target and never *detects* cartelists. Granular per-feature, per-fold, per-case, and calibration diagnostics, the full feature dictionary, and the supporting tables moved out of this appendix are deposited in Online Supplement E.

Appendix E.1. LANCES Bid-Layer Data, Units, and Support

The bid layer is built from `v3/data/processed/bid_level_with_prices.parquet` (approximately 245 MB), one row per priced bid with columns `códigofornecedor` (firm), `numerodaoc` (OC sequence), `códigoitem` (item), and `bid_price`; a bid is retained only if `bid_price IS NOT NULL AND bid_price > 0`. The benchmark is produced by script 31 (`31_imhof_full_pipeline.R`, `arrow + data.table`, seed 20260430) and a DuckDB twin, script 49 (`49_imhof_incremental_value.R`, seed 20260501); a dedicated validation run (`09_bid_benchmark_validation.R`, seed 20260603) generates the audit numbers. The bid layer links to the award layer through the tender-item key (`numerodaoc, códigoitem`) and, for firm attachment, through (`numerodaoc, códigoitem, códigofornecedor`); the within-tender moments are *not* recoverable from the award layer, which records no per-bid prices.

The modeling unit is the *firm*: per-tender moments are shared across all firms bidding on a tender-item and then firm-mean aggregated. Of the 16,843 always-losers, 64 are dropped because they have no priced bids or fewer than two bids per tender (so the coefficient of variation, skewness, and kurtosis are undefined). Excluding the direct CADE defendants by upstream label curation leaves an evaluation pool of 16,731 firms, of which 651 are always-loser cobidder positives. **All 651 positive (cobidder) labels are non-missing on every Imhof moment**: the support loss falls entirely on the negatives, so there is no positive attrition. (The full units-of-observation and per-feature missingness tables are in Online Supplement E.)

Appendix E.2. Bid-Feature Families

Seven Imhof–Wallimann moments are constructed within each tender-item with ≥ 2 priced bids, then averaged across each firm’s priced tenders to the firm level. They fall

into four families: *dispersion* (coefficient of variation `imhof_cv_mean` and its within-firm SD `imhof_cv_sd`, undefined for single-tender firms); *level/relative spread* (normalized spread and log min–max, `imhof_spread_mean`, `imhof_minmax_mean`); *shape* (skewness `imhof_skew_mean` and excess kurtosis `imhof_kurt_mean`); and *rank/order* (the second-low gap $\log(b_2/b_1)$, `imhof_second_low_mean`, the only moment with a winning-margin component, of low–moderate leakage risk but still a field-level moment shared across firms, not a firm-specific outcome). Two auxiliary counts accompany them, `n_bids` (gating the ≥ 2 filter) and `n_tenders_priced` (a firm exposure count). No co-bidding/network or timing, bid-revision, or late-bid features are implemented: the parquet read selects only firm, OC, item, and price, and no timestamp column is touched. The full feature dictionary with all columns and formulas is in Online Supplement E.

Appendix E.3. Model Specification

The learner is a random forest (`ranger`, 500 trees, `probability = TRUE`, `num.threads = 12`) — **not** a logit, despite an earlier script header that mislabeled it. There is no hyperparameter tuning (fixed 500 trees, default `mtry` and `min.node.size`, no grid search) and no standardization (the forest is scale-invariant). Missingness is handled listwise through the pool restriction. The outcome is binary `is_cade` (1 for an always-loser cobidder in the candidate pool, not a direct defendant). Evaluation is by 5-fold cross-validation, each firm receiving one held-out predicted probability; AUC and DeLong tests are computed on these out-of-fold firm-level predictions. (The full spec-by-spec reproduction audit is in Online Supplement E.)

Appendix E.4. Cross-Validation and Holdout Designs

Four validation designs are run, increasing in stringency: **Design A** (pooled-random 5-fold) is a diagnostic *upper bound*; **Design E** (case-grouped / leave-one-case-out) holds out an entire CADE case so its clustered positives never co-train; **Design F** (exclude-label-defining tenders) recomputes the moments after dropping defendant-co-appearance tenders to test feature–target contamination; and **Design G** (environment holdout) holds out an item-group. Strict bid-timing validation is **blocked**: the within-tender moments cannot be cleanly time-limited without re-deriving them on a per-period bid panel. Because positives cluster by case, random folds (Design A) are optimistic —

same-case positives appear in both train and test — and the bid ROC drops from 0.717 (A) to 0.626 (E) while its PR-AUC falls from 0.116 to 0.062, against an award score that is identical across designs by construction. (Full fold-composition and per-case tables are in Online Supplement E.)

Appendix E.5. Leakage, Contamination, and Performance

The award score is *clean* (a CADE-label-independent rank on `log1p(tenders_count)`), as is its strict-timing variant. The bid and combined forests under full-period features and random folds are **diagnostic-only**: random folds are optimistic, and the firm-mean moments include label-defining tenders. The contamination is small: excluding label-defining tenders (Design F) leaves the bid ROC essentially unchanged ($0.717 \rightarrow 0.713$), so the fragility is case clustering, not feature–target overlap. The headline ranking performance is in Table E.1. On the pooled full-observability diagnostic the cheap award layer is *comparable* to the costly bid benchmark: award-continuous ROC 0.760 / PR-AUC 0.143 versus bid-RF ROC 0.717 / PR-AUC 0.116. The combined model is a full-observability *upper bound* (ROC 0.756 / PR-AUC 0.188 pooled), not a first-stage rule. Under case-grouped folds the bid layer falls hardest — bid-RF PR-AUC $0.116 \rightarrow 0.062$ — and, critically, the combined model’s PR-AUC falls *below* award-only (0.103 versus 0.143): the apparent complementarity does not survive case-grouped evaluation. The award and bid scores are only moderately correlated (Spearman ≈ 0.544 overall), so the two layers order firms differently; adding the award signal to the bid forest raises PR-AUC by +0.072 under pooled-random folds (DeLong $p < 0.001$), but this increment is conditional on this implemented benchmark and is case-fragile. (The full leakage-audit, complementarity, and calibration tables are in Online Supplement E.)

Appendix E.6. Limitations

The bid-layer benchmark is a full-observability diagnostic, not a first stage: the Imhof moments require all within-tender bids, which is exactly the information cost the award-layer screen avoids; the headline pooled-random numbers carry random-CV optimism; strict bid-timing validation is *blocked*, because within-tender dispersion moments cannot be cleanly restricted to a pre-target window; the benchmark is deliberately modest (untuned,

Table E.1: Bid-layer performance by validation design.

Design	Model	ROC-AUC	PR-AUC	Prec@500	Lift@500
A (random)	Award (cont.)	0.760	0.143	0.216	5.55
A (random)	Award FL14	0.688	0.098	0.130	3.34
A (random)	Bid RF	0.717	0.116	0.180	4.63
A (random)	Combined RF	0.756	0.188	0.274	7.04
E (case-grouped)	Award (cont.)	0.760	0.143	0.216	5.55
E (case-grouped)	Award FL14	0.688	0.098	0.130	3.34
E (case-grouped)	Bid RF	0.626	0.062	0.086	2.21
E (case-grouped)	Combined RF	0.689	0.103	0.190	4.88
F (excl. label)	Award (cont.)	0.829	0.156	0.216	6.33
F (excl. label)	Bid RF	0.713	0.088	0.130	3.81
F (excl. label)	Combined RF	0.855	0.217	0.302	8.85
G (env. holdout)	Award (cont.)	0.768	0.185	0.150	2.76
G (env. holdout)	Bid RF	0.747	0.171	0.150	2.76
G (env. holdout)	Combined RF	0.734	0.212	0.144	2.65

$N = 16,731$ always-losers with 651 always-loser cobidder positives (defendant-excluded pool) for Designs A, E; $N = 16,665$ with 569 positives for Design F; $N = 2,377$ with 129 positives for the held-out environment in Design G. PR-AUC is reported alongside ROC because the positive base rate is near one percent. Design A is a full-observability *diagnostic upper bound* with random-CV optimism; the case-grouped block (E) is the honest robustness test, where the combined model’s PR-AUC (0.103) falls below award-only (0.143). The award rank is identical across A and E by construction. Cobidders are adjudication-anchored exposure, not cartel membership; false positives are unlabeled firms, not “bad firms.”

unstandardized Imhof–Wallimann moments in a random forest), not a state-of-the-art horse race; and the target is the 651 always-loser cobidders, an adjudication-anchored exposure measure, so false positives are unlabeled firms and nothing here constitutes proof of conduct.

Appendix E.7. Sequential Gatekeeping and the Cost-Recall Frontier

The sequential architecture in the main text separates two tasks that are often conflated. The award-layer stage ranks firms using fields available before any bid microdata are recovered; the bid-layer stage then constructs richer forensic features only for the firms that survive the first stage. This appendix states the sequence, measures the recovery footprint each stage opens, and traces the cost-recall frontier that results, so the empirical comparison reads as an evidence-allocation design rather than a claim that thin records substitute for full bid files. The output is a forensic-priority queue, not a liability finding. Throughout, “cost” denotes a *data-processing footprint* — the firms, tender-items, bid rows, and buyer-item-year cells that must be opened to compute the bid-layer features — and never a monetary agency cost, which we do not observe. Discriminating performance is adjudication-anchored: positives are the 651 always-loser cobidder firms that link to a CADE-adjudicated case, against the complete-Imhof pool of 16,772 firms. Throughout,

the operating points should be read as retrospective cost-footprint designs conditional on the validated incumbent ranking, not as fully prospective deployment tests.

Appendix E.7.1. Candidate Pools and Ranking Rules

The evaluation pool (Pool A) is the complete-Imhof firm set: 16,772 firms, of which 651 are adjudication-anchored positives; under full observability it opens 116,112 distinct tender-items and 3,393,915 within-tender bid rows. Three prediction scores are defined over this pool: an **award score**, the zero-win-stratum continuous participation score $s_i = \log(1 + T_i)$, computable from routine administrative fields with no bid values; a **bid score**, a random forest on within-tender bid-distribution moments requiring the full bid record of each opened tender-item; and a **combined score** on the union of the two feature sets. We compare seven ranking rules distinguished by which layer they consume and whether they triage before opening bid microdata: award-only (cheap, no bid recovery; the main-text screen); bid-only and joint full-observability (both require opening every tender-item up front; the joint rule is an upper-bound comparison, not an operational baseline); sequential award $\rightarrow$ bid and award $\rightarrow$ combined (rank by award, keep the top- K_1 survivor pool, recover bid microdata only inside it, then rerank survivors); FL14 $\rightarrow$ bid (a binary-cutoff first-stage triage followed by bid reranking); and a random K_1 -sized survivor pool, the no-information floor. An opportunity-adjusted award score cannot serve as a gate — the exposure adjustment requires the same tender-level coverage the bid layer recovers — so it is reported only as an in-sample diagnostic in the main text. For the case-holdout exercise firms are grouped by the CADE case they link to, so held-out positives share no case with the calibration set.

Appendix E.7.2. Cost Denominators

Bid-layer recovery is costlier at scale because within-tender moments require the *full* set of bids in each opened tender-item, not only the focal firm’s bids: opening a survivor firm means opening every tender-item that firm participated in, in full. We therefore measure the recovery footprint along several denominators rather than a single headline number. Table E.2 defines them.

The firm count (D1) and the bid-row footprint (D4) diverge sharply, and the divergence is the central caution of this appendix. Survivors are high-participation firms: at the $K_1 = 2000$ operating point the survivor pool is about 12% of the firms (2,000/16,772, an

Table E.2: Cost denominators for the recovery footprint.

ID	Denominator (full-observability value)	Value
D1	Firms opened ($= \mathcal{S} $)	16,772
D2	Survivor firm-tender-item rows (with multiplicity)	163,264
D3	Opened tender-items (de-duplicated; recovery unit)	116,112
D4	Opened bid rows, full-tender (processing footprint)	3,393,915
D5	Survivor bid rows only (lower bound)	391,610
D6	Buyer $\times$ item-group $\times$ year cells touched	58,936
D7	Buyers (PBU codes) touched	1,244
D8	Item-groups (2-digit) touched	90

Source: `full_observability_costs.csv`; denominator definitions in `table_G_cost_denominator_definitions.csv`. Values are the full-pool footprint ($K_1 = 16,772$); the grid in [Appendix E.7.3](#) reports them at each K_1 .

D4 is the realistic processing footprint: within-tender Imhof moments require the entire tender record, so D4 counts all bids in every opened tender-item, not only the survivors' own bids (D5).

Not observed (no value computed): LANCES export units (assumed one tender-item = one export, D9); legal or subpoena-style access requests (D10); monetary cost in reais.

Analyst-hours are reported only as an illustrative processing-burden proxy ($\alpha \cdot \text{tender-items} + \beta \cdot \text{bid-rows}/1000$, low/medium/high), never as a measured agency cost (D11).

88% firm reduction), yet it opens about 67% of the bid rows ($2,283,924/3,393,915$, only a 33% bid-row reduction). The firm-level reduction therefore overstates the saving; the tender-item and bid-row denominators are the honest processing-footprint measures.

Appendix E.7.3. Cost-Recall Grid and Frontier

Table [E.3](#) reports a compact slice of the cost-recall grid for the sequential award $\rightarrow$ bid rule at the $k = 500$ forensic-queue depth, across the first-stage cut K_1 . The full grid — all rules, all K_1 , and $k \in \{100, 250, 500, 1000\}$, with every cost denominator — is in the online supplement.

Figure [E.1](#) plots recall against the tender-item footprint for the award-only, sequential, and joint rules, making the frontier shape visible: the sequential rule tracks close to the joint upper bound while opening fewer tender-items, but never crosses it.

Appendix E.7.4. Baselines, Timing, and Case-Holdout

The sequential rule sits well above a same-size random survivor floor across the grid, confirming that it is the award ranking, not the mere act of opening K_1 firms, that carries the signal. The award-ranked survivor pool captures most of the 651 positives *before* any bid rerank, so the cheap first stage alone already gathers the positives into the survivor pool, and the bid rerank then reorders that pool into a forensic-priority queue rather than

Table E.3: Cost-recall slice, sequential award $\rightarrow$ bid, queue depth $k = 500$.

K_1	Firms opened	Bid rows opened	Bid-row red.	TP	Recall
500	500	1,356,346	0.60	108	0.166
1,000	1,000	1,783,750	0.47	124	0.190
2,000	2,000	2,283,924	0.33	116	0.178
3,000	3,000	2,575,274	0.24	115	0.177
5,000	5,000	2,921,977	0.14	102	0.157
16,772	16,772	3,393,915	0.00	102	0.157

Source: `table_6_cost_recall_frontier.csv` (sequential award $\rightarrow$ bid, $k = 500$). “Bid-row red.” is the fraction of the full-pool bid-row footprint avoided, $1 - (\text{bid rows opened})/3,393,915$. TP counts adjudication-anchored positives in the top- k queue (pool = 651).

TP is non-monotone in K_1 because the bid rerank reorders survivors within a fixed queue depth: $K_1 = 1,000$ recovers more true positives (124) than $K_1 = 2,000$ (116) at $k = 500$, one more sign that no K_1 is optimal. The joint full-observability TP at $k = 500$ is 133, which the sequential rule approaches ($124/133 = 93\%$ at $K_1 = 1,000$) but does not exceed.

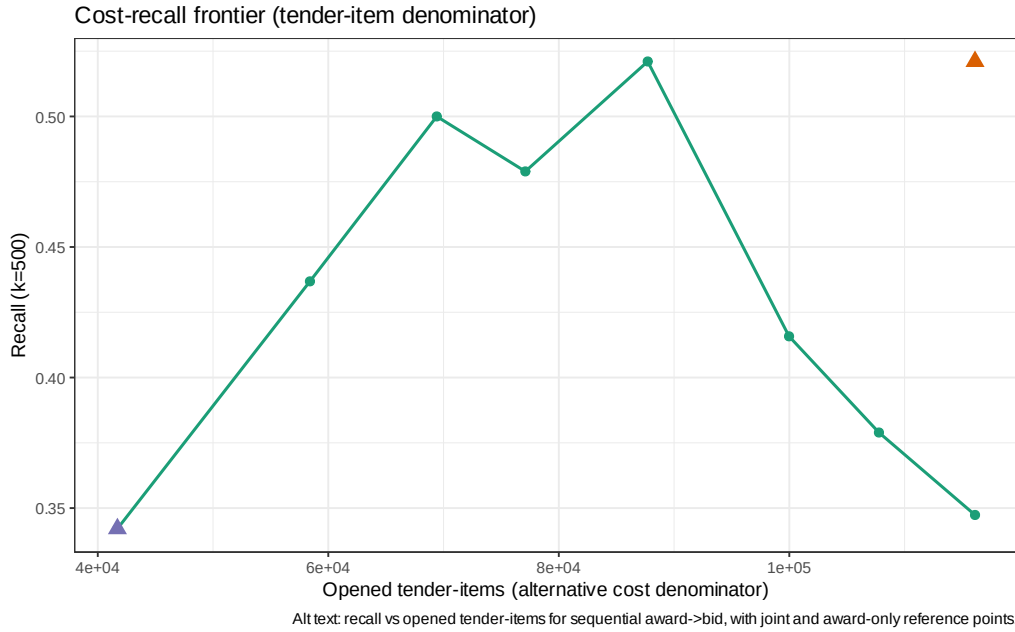


Figure E.1: Recall versus opened tender-items, by ranking rule.

Alt text: A cost-recall scatter with opened tender-items on the horizontal axis and adjudication-anchored recall on the vertical axis. The joint full-observability rule sits at the far right (largest footprint) at the top of the recall range; award-only points sit at the far left (smallest footprint) at lower recall; sequential award $\rightarrow$ bid points trace an intermediate frontier that approaches but does not exceed the joint recall while opening fewer tender-items.

recovering positives the award gate missed. This is the precise sense in which the bid layer reorders rather than discovers: the marginal contribution of the costly bid features is ordering, an evidence-allocation gain, not a detection gain.

A timing-disciplined gate must rank on information available before the investigation window. Re-scoring the award gate on a strict pre-period (2009–2016) and evaluating on later positives, the award gate remains informative under this discipline: its strict-timing discrimination is AUC 0.684 on the training-AL pool, against 0.761 for the full-window score. The drop is the honest cost of respecting the timing order. The sequential rule cannot be subjected to the same test — the within-tender moments cannot be re-derived on a pre-period basis without re-pipelining the raw LANCES bid records — so we report it as infeasible rather than estimate it on contaminated features.

Because adjudicated positives cluster in a handful of CADE cases, apparent recall can be carried by a single large case: the largest case supplies 32.0% of the positives and 45.4% of the true positives in the top-500 queue. Grouping firms by case and holding out the largest case lowers recall on the remaining smaller cases materially, with several individual small cases near zero. Performance is therefore case-fragile and generalization to out-of-sample cartels is weaker than the pooled figure suggests. The estimated ranking is not a portable cartel score; the transferable object is the decomposition framework. (The full random and award-only baseline table and the case-holdout grid are in Online Supplement E.)

Appendix E.7.5. Operating-Point Logic

No single (K_1, k) pair is preferred for every agency; the right point depends on capacity and on whether the priority is recall or precision — the frontier is a menu, not a cutoff. At the $k = 500$ queue depth the frontier spans a capacity-constrained low end ($K_1 = 500$, smallest footprint 1,356,346 bid rows, recall 0.166, precision 0.216) through $K_1 = 1,000$ (recall 0.190, precision 0.248) and $K_1 = 2,000$ (recall 0.178, precision 0.232) to $K_1 = 3,000$ (recall 0.177, precision 0.230). Recovered positives are non-monotone in K_1 : $K_1 = 1,000$ beats $K_1 = 2,000$ (124 versus 116 true positives), one more sign that no K_1 is optimal. The $K_1 = 2,000$ point used in the main text for the firm-versus-bid-row divergence is one operating point on this frontier, not a universally preferred choice. The full operating-points table with use-case labels, and the cost-per-true-positive and

false-positive frontier figures, are in Online Supplement E.

Appendix E.7.6. Algorithm

The full gatekeeping procedure can be stated compactly.

Algorithm G.1: Award-Layer Gatekeeping with Cost Measurement

1. Define the firm universe from award records (participant identity, winner identity, item code, negotiated price, procuring unit, year, modality).
2. For each firm i , compute the win count W_i and the participation count T_i over the calibration window; restrict to the zero-win stratum $\{i : W_i = 0\}$.
3. Compute the award score $s_i = \log(1 + T_i)$.
4. Over a grid of first-stage cuts K_1 , form the survivor pool $\mathcal{S}_{K_1}$ of the top- K_1 firms (or apply the median-plus-1.5×IQR threshold, or a capacity budget).
5. Recover bid microdata only for the tender-items touched by $\mathcal{S}_{K_1}$, and record the cost denominators D1–D8 at that K_1 .
6. Compute the bid-layer forensic features inside $\mathcal{S}_{K_1}$.
7. Rerank survivors by the bid score alone or by the award-plus-bid combined score, over a grid of queue depths k .
8. Deliver the top- k queue from that ranking as a forensic-priority list for investigative review.
9. Trace the cost-recall frontier over (K_1, k) and select the operating point that fits the agency’s capacity and recall/precision priority; treat the output as forensic priority, not as a liability finding.

Joint scoring requires bid microdata for the full pool before triage; sequential gatekeeping requires bid microdata only after the award-layer survivor pool is formed. The size of the survivor pool, the choice of threshold or top- K_1 , the queue depth k , the bid-versus- combined rerank, and the recalibration cadence are all institutional choices that the algorithm leaves to the agency.

Appendix E.7.7. Limitations

Cost is a processing footprint, not money — we measure firms, tender-items, bid rows, and cells opened, not monetary recovery cost, legal-request volume, or LANCES export units. Firm reduction overstates the saving (an 88% firm reduction at $K_1 = 2000$ is only a 33% bid-row reduction); analyst-hours are illustrative only; labels are limited and false positives are unlabeled, so precision is a lower bound on the queue’s triage value; performance is case-fragile, with the largest case supplying 32.0% of positives and leave-largest-case-out recall falling materially; the operating point is capacity-dependent with no dominating choice; and the sequential timing test is blocked, so only the award-only gate has a timing-disciplined evaluation.

Appendix F. Price Scope and Adaptive Use

The main text (Section 7) treats price evidence as corroboration of economic non-neutrality, not as causal identification. This appendix records the price sample, the broad associations, the overlap-cell reweighting that reverses their sign, the high-value exception, the direct-CADE null, and the boundary diagnostics that keep the interpretation as scope rather than a price effect. Every number is the same object reported in the main text; the full regression grids live in Online Supplement G. The strategic-adaptation simulation is reported separately in [Appendix F.6](#).

Appendix F.1. Price sample and outcome

The outcome is the log negotiated unit price at the item–award level. The estimation sample is the broad item-level panel with $N = 1,654,401$ awards, with item, year, and (where indicated) procuring-unit (PBU) fixed effects. The overlap, quartile, and trim variants reweight or restrict this sample but do not change the outcome or the unit of observation.

Appendix F.2. Broad associations and overlap-cell sign reversal

In the broad specification, frequent-loser presence is positively associated with the log negotiated unit price, with a headline range of $+3.6\% - +7.7\%$ across estimators (broad coefficient $+0.064$; the larger association in the electronic-auction environment is

inconsistent with a pure quorum-filler reading). These are descriptive associations, not causal effects; the full broad grid by modality is in Online Supplement G.

The positive broad coefficient is a weighting artifact, not a robust treatment sign. Holding the overlap cells fixed and moving from broad-sample weighting to the within-cell average treatment effect on the treated (ATT) reverses the sign monotonically: broad $+0.064$ ($p = 0.003$); overlap-cell unweighted $+0.044$ ($p = 0.035$); overlap-cell ATT -0.097 ($p < 0.001$); PS-trimmed -0.307 . Only 1.06% of treated items lack a within-cell counterfactual, so the reversal cannot be a dropped-observation artifact. Table F.1 reports the progression.

Table F.1: Sign-Reversal Decomposition (overlap-cell reweighting)

Specification	Coefficient	p -value
Broad sample	$+0.064$	0.003
Overlap-cell, unweighted	$+0.044$	0.035
Overlap-cell ATT	-0.097	< 0.001
Overlap-cell ATT, PS-trimmed	-0.307	—

Notes: Same outcome and overlap cells throughout; rows differ only in weighting and trimming. The broad-to-ATT move reverses the sign because it strips the between-cell selection component. Only 1.06% of treated items lack a within-cell counterfactual.

Appendix F.3. High-value segment, direct-CADE null, and the scope boundary

The negative ATT reading survives both modalities (Convite -0.099 , Pregão -0.098) and every tender-value quartile except the highest; the high-value tail (Q4) is the only systematically positive cell ($+0.041$, $p = 0.045$), where coordination costs are spread over larger contracts. Trimming the most heavily weighted overlap cells does not eliminate the negative aggregate. The full quartile and trim grid is in Online Supplement G. Restricting the overlap to items involving firms named as direct CADE defendants yields a null price estimate (-0.061 , $p = 0.45$; non-direct overlap -0.097): the price regressions do not recover the legal object of the adjudicated cases. Three facts keep the evidence on the scope side of the line. The broad positive imprint is selection into structurally higher-priced cells (among non-treated items, mean log price rises from 1.35 to 6.93 across cell FL-share quintiles, a $\Delta = 5.58$ log-point gap, with a partialled-out coefficient $+3.55$ once all five cell dimensions are absorbed), and the within-cell ATT removes exactly this component. The direct-CADE null means there is no price base tied to the adjudicated

conduct. And the design holds the procurement environment fixed rather than against a counterfactual price, so no overcharge can be computed. *These are scope facts about where flagged environments sit in the price distribution, not damages or overcharge.*

Appendix F.4. Within-cell mechanism

The within-cell price compression loads on the count of genuine bidders, not on the frequent-loser count: the within-cell coefficient is -0.048 (SE 0.004) and falls to $+0.008$ (not significant) once $\log(n_firms)$ is added, so the mechanism is *not* identified by these data.

Appendix F.5. Limitations

The price evidence is descriptive, environment-specific, and not causal. It establishes economic non-neutrality of flagged environments, concentrated in the high-value tail, but it is not a damages estimate, an overcharge, a markup, or a treatment effect on price. The coefficients are BEC-specific; another agency should re-validate the ranking against its own enforcement anchors and price benchmarks rather than reusing these estimates or the BEC cutoff.

Appendix F.6. Adaptive Triage Designs (Retrospective)

The award-layer score supports an incumbent-firm triage design, so a diagnostic simulation holds the empirical environment fixed and asks how the participation score behaves under simple evasive responses. This is a retrospective robustness probe, not a fully prospective deployment test: the labels remain adjudication-anchored, and the operating points should be read as conditional on the validated incumbent ranking. Rotation of losing-role firms and occasional small wins leave discrimination essentially unchanged (0.730 and 0.720 versus baseline 0.722), absorbed by the persistence signal; CNPJ splitting lowers AUC to 0.653 by breaking participation history; threshold-aware capping is the most damaging (0.676 , and 0.607 combined with rotation). The hierarchy maps to the cost-recall frontier: under an incumbent-triage reading the rule should be refreshed as new participation data arrive, and continuous ranks or capacity-constrained top- k queues should be preferred over a static published cutoff, since a published threshold invites bunching (consistent with the local density ratio ≈ 1.06 at $T = 14$ in the profile appendix, where the threshold was computed ex post and never disclosed) and is the most

gameable rule. The summary table appears in Online Supplement G. This is a diagnostic, not a full equilibrium model of evasion: strategic response exposes the vulnerability of a static binary rule more than it invalidates the two-stage architecture, and none of these magnitudes should be read as a platform-wide prospective deployment result.